

```

1 global:
2   scrape_interval: 15s
3 scrape_configs:
4   - job_name: node
5     static_configs:
6       - targets: ['node-exporter:9100']
7   - job_name: "apache"
8     static_configs:
9       - targets: ['apache-exporter:9117']

```

1. คำสั่งสร้าง network monitoring

```

(base) PS C:\Users\USER> docker network create monitoring
158b204380f858c46be1d9e04fbb5cf4cceed7a45c90f974d3dbd83193132ce1
(base) PS C:\Users\USER> docker network connect monitoring grafana prometheus node-exporter
docker: 'docker network connect' requires 2 arguments

Usage: docker network connect [OPTIONS] NETWORK CONTAINER

Run 'docker network connect --help' for more information
(base) PS C:\Users\USER> docker network connect monitoring grafana
(base) PS C:\Users\USER> docker network connect monitoring prometheus
(base) PS C:\Users\USER> docker network connect monitoring node-exporter
(base) PS C:\Users\USER> docker network inspect monitoring

```

```

"Containers": {
  "059d1bd764003bd9dd5f56e60ebc6d635a90a145b4dee41e8ed4595429aa451b": {
    "Name": "grafana",
    "EndpointID": "730d1e11e81163961391382f07ccc19b3ba3bea6feef5a999524075c3ce730fd",
    "MacAddress": "0e:5a:7c:80:76:4a",
    "IPv4Address": "172.24.0.2/16",
    "IPv6Address": ""
  },
  "785d779c60a272218e46e21586598b0cf1d5bc0a3c6ad9bcae609ca7df145372": {
    "Name": "prometheus",
    "EndpointID": "d25153fb1d42933f042175beed74f2d502dc0292f2d62382ea137412a10adf2c",
    "MacAddress": "76:26:72:8c:1f:7f",
    "IPv4Address": "172.24.0.3/16",
    "IPv6Address": ""
  },
  "e94e7e8824c58e6afa218499c4bea85132c5b3acdc82d01044cec6eb0d8b7156": {
    "Name": "node-exporter",
    "EndpointID": "5d1aefe86f90a36534c52174a9e5d671c31077524e6ee7ff2799aa754bf7c662",
    "MacAddress": "de:de:9f:b6:b3:3f",
    "IPv4Address": "172.24.0.4/16",
    "IPv6Address": ""
  }
}

```

2. คำสั่งสร้าง persistent storage ของ volume Grafana และ bind mount status.conf volume Grafana:

```
(base) PS C:\Users\USER> docker run --rm -d -p 3000:3000 --network mynet -v grafana-vol:/var/lib/grafana --name grafana grafana/grafana:11.4.0
Unable to find image 'grafana/grafana:11.4.0' locally
11.4.0: Pulling from grafana/grafana
9b4925e32b92: Pull complete
8964b0551c55: Pull complete
989e41ffb5ca: Pull complete
da9db072f522: Pull complete
ebc122b722: Pull complete
d7013a437817: Pull complete
1739eae10c: Pull complete
b2ddcf85da57: Pull complete
e0862465767a: Pull complete
00a90dd48f56: Pull complete
Digest: sha256:d8ea37798ccc41061a62ab080f2676dda6bf7815558499f901bdb0f533a456fb
Status: Downloaded newer image for grafana/grafana:11.4.0
059d1bd764003bd9dd5f56e60ebc6d635a90a145b4dee41e8ed4595429aa451b
```

bind mount status.conf

```
(base) PS C:\Users\USER> docker run --rm -d --name apache --network monitoring -p 8080:80 -v C:\Users\USER\Desktop\Software_def\multiple_container\status.conf:/etc/apache2/mods-enabled/status.conf ubuntu/apache2:latest
b57f0344d19529d2c2c77ceceb435522d75e0372d0298982bed6dd418c725edf
```

3. คำสั่งรัน container แต่ละตัว

apache:

```
(base) PS C:\Users\USER> docker run --rm -d --name apache --network monitoring -p 8080:80 -v C:\Users\USER\Desktop\Software_def\multiple_container\status.conf:/etc/apache2/mods-enabled/status.conf ubuntu/apache2:latest
b57f0344d19529d2c2c77ceceb435522d75e0372d0298982bed6dd418c725edf
```

apache-exporter:

```
(base) PS C:\Users\USER> docker run --rm -d --name apache-exporter --network monitoring -p 9117:9117 lusotycoon/apache-exporter --scrape_uri="http://apache/server-status/auto"
a8379ca550c34da933f3ede792b9386a1c7e5fcb3dcf7faeab259868685c697e
```

Grafana:

```
(base) PS C:\Users\USER> docker run --rm -d -p 3000:3000 --network monitoring -v grafana-vol:/var/lib/grafana --name grafana grafana/grafana:11.4.0
7721fda28ddd7b827b00145e86343f284d0bfb887ddf38ae9e2ae56ee5a98c5
```

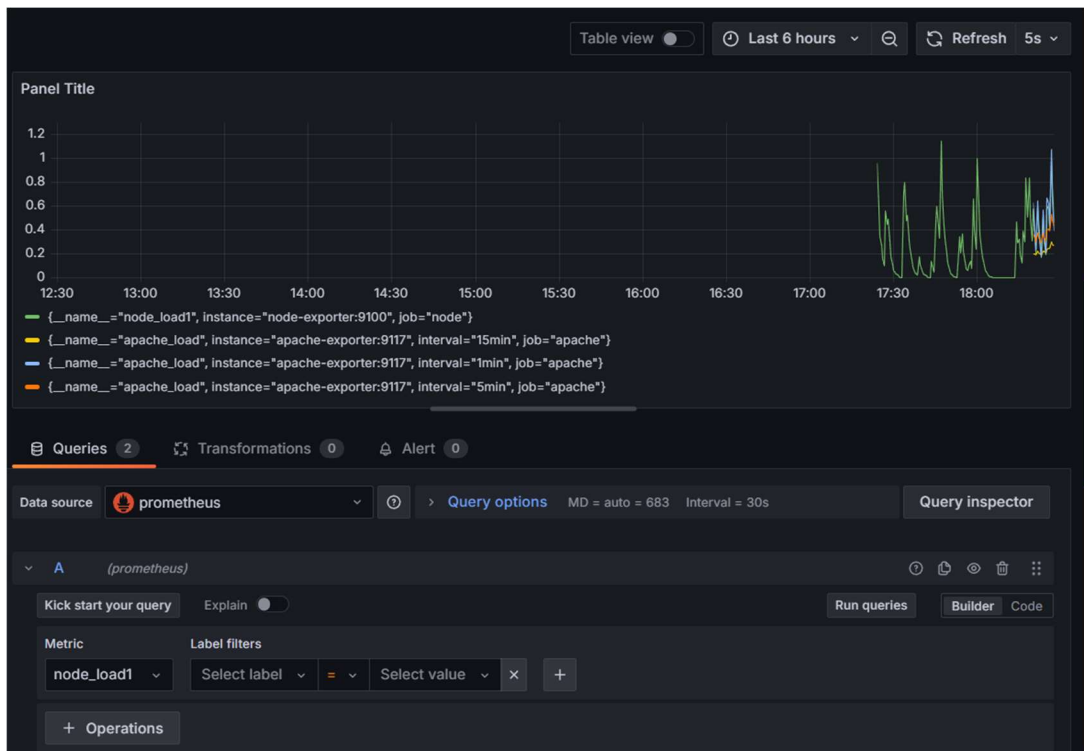
Prometheus: (มาเชื่อม network monitoring กับ rename ที่หลัง)

```
(base) PS C:\Users\USER> docker run --rm -d -p 9090:9090 --network mynet -v C:\Users\USER\Desktop\Software_def\multiple_container\prometheus.yml:/etc/prometheus/prometheus.yml prom/prometheus:v3.0.1
```

node-exporter: (มาเชื่อม network monitoring ที่หลัง)

```
(base) PS C:\Users\USER> docker run --rm -d -p 9100:9100 --network mynet --name node-exporter prom/node-exporter
```

Grafana dashboard



Node_load1 คือ load ที่ os ใช้ในช่วงเวลา 5 วินาที (กำหนดเป็น every 5 second) โดยจะดูทั้ง process หรือ thread หรือสิ่งที่กำลังรออยู่ใน queue ด้วย

apache_load1 คือ ภาระงานที่เกิดขึ้นจากการรัน apache container ซึ่งแบ่งเป็น 3 ช่วงเวลา คือ 15 min, 1 min, 5min